Black–Scholes from Replication

Lecture 5 Notes — Self-Financing, PDE, Verification, and Greeks

Goals and logical structure

This lecture connects stochastic calculus to option valuation through one reproducible chain:

Itô dynamics $\longrightarrow$ self-financing replication $\longrightarrow$ Black–Scholes PDE $\longrightarrow$ Gaussian expectation and Greeks.

We first practice Itô's formula on examples not used in Lecture 4. We then define self-financing carefully in discrete and continuous time, derive the PDE for a European payoff $g(S_T)$, solve and verify it, compute the call and put formulas and their Greeks, and conclude with the practical limits of continuous-time hedging.

1 Three New Uses of the Itô Formula

1.1 The generator organizes the calculation

Let

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + a(t, X_t)\,\mathrm{d}W_t$$

and $f \in C^{1,2}$. The one-dimensional Itô formula is

$$\mathrm{d}f(t, X_t) = \left(f_t + bf_x + \frac{1}{2}a^2 f_{xx}\right)(t, X_t)\,\mathrm{d}t + a(t, X_t)f_x(t, X_t)\,\mathrm{d}W_t.$$

It is convenient to define the generator

$$\mathcal{L}f(t, x) = b(t, x)f_x(t, x) + \frac{1}{2}a(t, x)^2 f_{xx}(t, x),$$

so that

$$\boxed{\mathrm{d}f(t, X_t) = \big(f_t + \mathcal{L}f\big)(t, X_t)\,\mathrm{d}t + a(t, X_t)f_x(t, X_t)\,\mathrm{d}W_t.}$$

The operator $f_t + \mathcal{L}f$ is exactly the drift of the transformed process.

Driftless is not automatically a true martingale

If $f_t + \mathcal{L}f = 0$, then $f(t, X_t)$ is first a local martingale. An integrability, boundedness, or localization argument is needed before taking expectations or conditional expectations.

1.2 Dimension creates drift

Example — the squared norm of Brownian motion

Let $W = (W^1, \ldots, W^d)$ be a standard d-dimensional Brownian motion and set

$$R_t = \|W_t\|^2 = \sum_{i=1}^{d} (W_t^i)^2.$$

Applying the multidimensional formula to $f(x) = \|x\|^2$ gives

$$\boxed{\mathrm{d}R_t = 2W_t^\top\,\mathrm{d}W_t + d\,\mathrm{d}t.}$$

Consequently,

$$R_t - dt = 2\int_0^t W_s^\top \, \mathrm{d}W_s$$

is a square-integrable martingale on bounded horizons, and

$$\mathbb{E}[\|W_t\|^2] = dt.$$

Each independent Brownian direction contributes one unit of quadratic variation per unit of time.

Exercise — replace the identity covariance

Suppose $X_t = CW_t$, where C is a deterministic $m \times d$ matrix. Find $\mathrm{d}\|X_t\|^2$ and $\mathbb{E}\|X_t\|^2$.

Short solution

Since $\mathrm{d}X_t = C\,\mathrm{d}W_t$,

$$\mathrm{d}[X^i, X^j]_t = (CC^\top)_{ij}\,\mathrm{d}t.$$

Therefore

$$\mathrm{d}\|X_t\|^2 = 2X_t^\top C\,\mathrm{d}W_t + \mathrm{tr}(CC^\top)\,\mathrm{d}t,$$

and

$$\boxed{\mathbb{E}\|X_t\|^2 = t\ \mathrm{tr}(CC^\top).}$$

1.3 Adding a path variable restores the Markov state

Define integrated Brownian motion

$$A_t = \int_0^t W_s\,\mathrm{d}s.$$

The pair (W_t, A_t) is a two-dimensional state process:

$$\mathrm{d}W_t = \mathrm{d}W_t, \qquad \mathrm{d}A_t = W_t\,\mathrm{d}t.$$

Although A_t depends on the entire past of W, augmenting the state by A_t makes the dynamics local again.

Example — a product with integrated Brownian motion

For $Y_t = W_t A_t$, the product rule gives

$$\mathrm{d}Y_t = A_t\,\mathrm{d}W_t + W_t\,\mathrm{d}A_t + \mathrm{d}[W, A]_t.$$

The process A has finite variation, so $[W, A] = 0$, and hence

$$\boxed{\mathrm{d}(W_t A_t) = A_t\,\mathrm{d}W_t + W_t^2\,\mathrm{d}t.}$$

Taking expectations,

$$\boxed{\mathbb{E}[W_t A_t] = \int_0^t \mathbb{E}[W_s^2]\,\mathrm{d}s = \frac{t^2}{2}.}$$

This agrees with the direct covariance calculation

$$\mathrm{Cov}\left(W_t, \int_0^t W_s\,\mathrm{d}s\right) = \int_0^t \min(t, s)\,\mathrm{d}s = \frac{t^2}{2}.$$

1.4 A future event becomes a backward-PDE martingale

Let N and φ denote the standard normal cdf and density. Fix $T > 0$ and define, for $t < T$,

$$u(t, x) = \mathrm{N}\left(\frac{x}{\sqrt{T - t}}\right).$$

Writing $\tau = T - t$ and $z = x/\sqrt{\tau}$,

$$u_x = \frac{\varphi(z)}{\sqrt{\tau}}, \qquad u_{xx} = -\frac{x\varphi(z)}{\tau^{3/2}}, \qquad u_t = \frac{x\varphi(z)}{2\tau^{3/2}}.$$

Therefore

$$u_t + \frac{1}{2}u_{xx} = 0$$

and

$$\mathrm{d}u(t, W_t) = \frac{\varphi(W_t/\sqrt{T-t})}{\sqrt{T-t}}\,\mathrm{d}W_t.$$

The probabilistic meaning is

$$\boxed{u(t, W_t) = \mathbb{P}(W_T > 0 \mid \mathcal{F}_t).}$$

Indeed, conditional on $\mathcal{F}_t$,

$$W_T = W_t + \sqrt{T-t}\,Z, \qquad Z \sim N(0,1).$$

Preview of Black–Scholes

A terminal payoff, a backward PDE, a conditional expectation, and a replicating stochastic integral will again describe the same process.

2 Securities, Trading Strategies, and Self-Financing

2.1 Model assumptions and market assumptions

On a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0\le t\le T}, \mathbb{P})$, consider one risky asset and one bank account:

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t, \qquad B_0 = 1.$$

Thus $B_t = e^{rt}$. Throughout the basic Black–Scholes model, $\sigma > 0$.

Model assumptions

- The stock has continuous geometric Brownian dynamics.
- The parameters μ, σ, r are constant.
- The stock pays no dividends and does not default.
- One Brownian motion generates all uncertainty in the model.

Market assumptions

- Trading is continuous and frictionless.
- There are no transaction costs, bid–ask spreads, or taxes.
- Arbitrary long and short positions are allowed.
- Cash can be borrowed or lent at the same rate r.
- Strategies are predictable and admissible.
- The market satisfies no arbitrage.

Keep the logical roles separate

GBM is a model assumption. No arbitrage is a market principle. Completeness comes from having enough traded assets to span the modeled shocks. Constant coefficients and zero dividends are simplifications that produce the familiar closed form. None of these statements implies all the others.

2.2 Portfolio value and predictability

Definition 2.1 (Trading strategy)**.** A strategy consists of predictable processes (Δ_t, β_t), where Δ_t is the number of stock shares and β_t is the number of bank-account units held at time t. Its value is

$$\boxed{X_t = \Delta_t S_t + \beta_t B_t.}$$

Predictability expresses causality: the position used over the next instant must be selected from information already available. Suitable admissibility conditions also exclude doubling strategies and ensure that the stochastic integrals are well defined.

2.3 Self-financing in discrete time

The discrete formulation makes the cash-flow accounting transparent. Let $0 = t_0 < \cdots < t_N = T$. The holdings (Δ_n, β_n), chosen at time t_n, are kept over $(t_n, t_{n+1}]$.

Immediately before rebalancing at t_n, the old positions are worth

$$X_n^- = \Delta_{n-1} S_{t_n} + \beta_{n-1} B_{t_n}.$$

Immediately after rebalancing, the new positions are worth

$$X_n = \Delta_n S_{t_n} + \beta_n B_{t_n}.$$

Definition 2.2 (Discrete self-financing condition)**.** No cash is added or withdrawn at t_n:

$$X_n = X_n^-.$$

Equivalently,

$$\boxed{S_{t_n}(\Delta_n - \Delta_{n-1}) + B_{t_n}(\beta_n - \beta_{n-1}) = 0.}$$

Buying stock is financed by reducing the cash position; selling stock generates cash. Rebalancing changes composition, not wealth.

Between t_n and t_{n+1}, the assets move while the holdings remain fixed. Hence

$$\boxed{X_{n+1}^- - X_n = \Delta_n(S_{t_{n+1}} - S_{t_n}) + \beta_n(B_{t_{n+1}} - B_{t_n}).}$$

Portfolio gains come from asset-price changes, not from the act of rebalancing.

2.4 Continuous-time self-financing

Definition 2.3 (Continuous self-financing condition)**.** A strategy is self-financing when

$$\boxed{X_t = X_0 + \int_0^t \Delta_u \, \mathrm{d}S_u + \int_0^t \beta_u \, \mathrm{d}B_u.}$$

In differential shorthand,

$$\boxed{\mathrm{d}X_t = \Delta_t \, \mathrm{d}S_t + \beta_t \, \mathrm{d}B_t.}$$

The stochastic integrals are limits of predictable left-point trading gains. Using

$$\beta_t B_t = X_t - \Delta_t S_t, \qquad \mathrm{d}B_t = r B_t \, \mathrm{d}t,$$

the gain equation becomes

$$\boxed{\mathrm{d}X_t = \Delta_t \, \mathrm{d}S_t + r(X_t - \Delta_t S_t) \, \mathrm{d}t.}$$

Proposition 2.4 (Equivalent rebalancing constraint)**.** If Δ is a continuous semimartingale and $X_t = \Delta_t S_t + \beta_t B_t$, then self-financing is equivalent to

$$\boxed{S_t \, \mathrm{d}\Delta_t + \, \mathrm{d}[\Delta, S]_t + B_t \, \mathrm{d}\beta_t = 0.}$$

Proof. The product rule gives

$$\mathrm{d}X_t = \Delta_t\,\mathrm{d}S_t + S_t\,\mathrm{d}\Delta_t + \mathrm{d}[\Delta, S]_t + \beta_t\,\mathrm{d}B_t + B_t\,\mathrm{d}\beta_t.$$

Comparing this with the self-financing gain equation yields the claim. □

Value identity versus gain identity

The identity $X = \Delta S + \beta B$ says what the portfolio is worth. The identity $\mathrm{d}X = \Delta\,\mathrm{d}S + \beta\,\mathrm{d}B$ says how its value changes without external cash. A dynamic hedge needs both statements.

3 Replication and the Black–Scholes PDE

3.1 The European claim and its Itô dynamics

Fix a European payoff

$$H = g(S_T).$$

Assume that its value can be written as a sufficiently smooth Markov function

$$V_t = p(t, S_t), \qquad p(T, s) = g(s).$$

The condition $p \in C^{1,2}([0,T) \times (0,\infty))$ will be used before maturity. The payoff itself may have a kink at T.
By Itô's formula,

$$\boxed{\begin{aligned}\mathrm{d}V_t = &\left(p_t + \mu S_t p_s + \frac{1}{2}\sigma^2 S_t^2 p_{ss}\right)(t, S_t)\,\mathrm{d}t \\ &+ \sigma S_t p_s(t, S_t)\,\mathrm{d}W_t.\end{aligned}}$$

3.2 Match a self-financing replicating portfolio

A replicating strategy satisfies

$$X_t = p(t, S_t) \quad \text{for all } t, \qquad X_T = g(S_T).$$

Its self-financing dynamics are

$$\begin{aligned}\mathrm{d}X_t &= \Delta_t\,\mathrm{d}S_t + r(X_t - \Delta_t S_t)\,\mathrm{d}t \\ &= [rX_t + (\mu - r)S_t\Delta_t]\,\mathrm{d}t + \sigma S_t \Delta_t\,\mathrm{d}W_t.\end{aligned}$$

Matching the Brownian coefficients in $\mathrm{d}X_t = \mathrm{d}V_t$ gives

$$\boxed{\Delta_t = p_s(t, S_t).}$$

The stock holding in the replicating strategy is the option delta.
Now match the drift coefficients with $X_t = p(t, S_t)$:

$$rp + (\mu - r)sp_s = p_t + \mu s p_s + \frac{1}{2}\sigma^2 s^2 p_{ss}.$$

The physical drift cancels. We obtain the pricing equation.

Theorem 3.1 (Black–Scholes PDE for a European payoff)**.** Under the preceding model, market, regularity, and replication assumptions, the value p satisfies

$$\boxed{\begin{cases} p_t(t,s) + rs\,p_s(t,s) + \dfrac{1}{2}\sigma^2 s^2 p_{ss}(t,s) - rp(t,s) = 0, \quad t < T,\ s > 0, \\ p(T,s) = g(s). \end{cases}}$$

The replicating holdings are

$$\boxed{\Delta_t = p_s(t, S_t), \qquad \beta_t = \frac{p(t, S_t) - S_t p_s(t, S_t)}{B_t}.}$$

Why μ disappears

The option and the replicating portfolio have the same exposure to the stock. Their physical expected-return terms are therefore identical and cancel. Valuation depends on the cost of financing r and the unhedgeable quadratic-variation scale σ, not on an estimate of μ.

What the cancellation does not mean

The physical drift remains relevant for forecasting, portfolio choice, risk management, and the empirical distribution of hedging errors. It disappears from the replication price inside the idealized complete market.

3.3 No arbitrage and completeness

The replication argument uses two distinct economic ideas:

- **Completeness:** the one traded stock spans the one Brownian source of risk, so the option's diffusion coefficient can be matched.
- **No arbitrage:** a claim and a self-financing portfolio with the same terminal payoff must have the same initial price.

With jumps or an additional stochastic-volatility shock, one stock generally cannot match every source of uncertainty. The market may become incomplete, and no-arbitrage alone may determine a range of prices rather than a unique one.

Exercise — add a continuous dividend yield

Suppose the stock pays a continuous dividend yield q. A stock holder receives $qS_t\,\mathrm{d}t$ in addition to the price gain $\mathrm{d}S_t$. Derive the PDE for a European claim.

Short solution

The stock position earns

$$\Delta_t(\,\mathrm{d}S_t + qS_t\,\mathrm{d}t).$$

Repeating the self-financing replication argument gives

$$\boxed{p_t + (r-q)sp_s + \frac{1}{2}\sigma^2 s^2 p_{ss} - rp = 0, \qquad p(T,s) = g(s).}$$

The no-dividend model is the special case $q = 0$.

4 Solving the Black–Scholes PDE

4.1 Step 1: time to maturity

We solve

$$p_t + rsp_s + \frac{1}{2}\sigma^2 s^2 p_{ss} - rp = 0, \qquad p(T,s) = g(s).$$

Let

$$\tau = T - t, \qquad v(\tau, s) = p(T-\tau, s).$$

Since $p_t = -v_\tau$,

$$\boxed{v_\tau = rsv_s + \frac{1}{2}\sigma^2 s^2 v_{ss} - rv, \qquad v(0,s) = g(s).}$$

4.2 Step 2: remove the discounting term

Define

$$u(\tau, s) = e^{r\tau} v(\tau, s).$$

Then

$$u_\tau = ru + e^{r\tau} v_\tau, \qquad u_s = e^{r\tau} v_s, \qquad u_{ss} = e^{r\tau} v_{ss}.$$

The two ru terms cancel:

$$\boxed{u_\tau = rsu_s + \frac{1}{2}\sigma^2 s^2 u_{ss}, \qquad u(0,s) = g(s).}$$

4.3 Step 3: logarithmic coordinates

Set

$$x = \log s, \qquad w(\tau, x) = u(\tau, e^x).$$

The derivatives satisfy

$$w_x = su_s, \qquad w_{xx} = su_s + s^2 u_{ss},$$

and therefore

$$u_s = \frac{w_x}{s}, \qquad u_{ss} = \frac{w_{xx} - w_x}{s^2}.$$

Substitution yields

$$\boxed{w_\tau = \frac{1}{2}\sigma^2 w_{xx} + \left(r - \frac{1}{2}\sigma^2\right) w_x, \qquad w(0,x) = g(e^x).}$$

The coefficient $r - \sigma^2/2$ is the drift of the log price under the pricing dynamics.

4.4 Step 4: remove transport and solve the heat equation

Let

$$a = r - \frac{1}{2}\sigma^2, \qquad \varphi(\tau, y) = w(\tau, y - a\tau).$$

Then

$$\varphi_\tau = w_\tau - aw_x, \qquad \varphi_y = w_x, \qquad \varphi_{yy} = w_{xx},$$

so

$$\boxed{\varphi_\tau = \frac{1}{2}\sigma^2 \varphi_{yy}, \qquad \varphi(0,y) = g(e^y).}$$

If $Z \sim N(0,1)$, the heat-kernel solution is

$$\varphi(\tau, y) = \mathbb{E}\left[g\left(e^{y + \sigma\sqrt{\tau}Z}\right)\right].$$

Equivalently,

$$\varphi(\tau, y) = \int_{\mathbb{R}} g(e^z) \frac{\exp\left(-\frac{(z-y)^2}{2\sigma^2\tau}\right)}{\sqrt{2\pi\sigma^2\tau}}\, \mathrm{d}z.$$

Theorem 4.1 (General European-payoff formula)**.** Let $\tau = T - t$. Under suitable growth and regularity assumptions,

$$\boxed{p(t,s) = e^{-r\tau}\mathbb{E}\left[g\left(s\exp\left\{\left(r - \frac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}Z\right\}\right)\right], \qquad Z \sim N(0,1).}$$

Remark 4.2. This is a general formula, but not necessarily an elementary closed form. For an arbitrary payoff g, it reduces valuation to a one-dimensional Gaussian integral.

Regularity at maturity

Calls, puts, and digital payoffs are not C^2 at maturity. Gaussian convolution smooths their prices for every $t < T$. A fully general treatment distinguishes classical, weak, and viscosity solutions; here we use the classical formula before maturity and recover the payoff in the terminal limit.

Exercise — price a power payoff

For $g(s) = s^m$, with $m \in \mathbb{R}$ such that the expectation is finite, compute $p(t, s)$.

Short solution

Using $\mathbb{E}[e^{cZ}] = e^{c^2/2}$,

$$\begin{aligned} p(t,s) &= e^{-r\tau} s^m \exp\left(m(r - \tfrac{1}{2}\sigma^2)\tau + \tfrac{1}{2}m^2\sigma^2\tau\right) \\ &= \boxed{s^m \exp\left((m-1)r\tau + \frac{1}{2}m(m-1)\sigma^2\tau\right).} \end{aligned}$$

The cases $m = 0$ and $m = 1$ give a zero-coupon payment and one share, respectively.

5 Verification: PDE, Expectation, and Hedge

5.1 The discounted candidate must be a martingale

Introduce an auxiliary stock, started at $S_t^r = s$, with dynamics

$$\frac{\mathrm{d}S_u^r}{S_u^r} = r\,\mathrm{d}u + \sigma\,\mathrm{d}W_u, \qquad u \in [t, T].$$

Its terminal value is

$$S_T^r = s \exp\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma(W_T - W_t)\right).$$

Theorem 5.1 (Verification theorem)**.** Suppose $p \in C^{1,2}$ solves the Black–Scholes PDE with terminal condition g, and assume the stochastic integral below is a true martingale. Then

$$\boxed{p(t,s) = \mathbb{E}\left[e^{-r(T-t)} g(S_T^r) \mid S_t^r = s\right].}$$

Moreover, in the original market, the strategy

$$\Delta_t = p_s(t, S_t), \qquad \beta_t = \frac{p(t, S_t) - S_t p_s(t, S_t)}{B_t}$$

is self-financing and replicates $g(S_T)$.

Proof. Apply Itô's formula and the product rule:

$$\begin{aligned} \mathrm{d}\left(e^{-r(u-t)} p(u, S_u^r)\right) &= e^{-r(u-t)}\left[p_t + rS_u^r p_s + \frac{1}{2}\sigma^2 (S_u^r)^2 p_{ss} - rp\right]\mathrm{d}u \\ &\quad + e^{-r(u-t)}\sigma S_u^r p_s\,\mathrm{d}W_u. \end{aligned}$$

The drift vanishes by the PDE. Taking conditional expectations between t and T and using $p(T, \cdot) = g$ proves the representation.

For replication under the original dynamics, rearrange the PDE inside Itô's formula:

$$\mathrm{d}p(t, S_t) = p_s(t, S_t)\,\mathrm{d}S_t + r\big(p(t, S_t) - S_t p_s(t, S_t)\big)\,\mathrm{d}t.$$

This equals $\Delta_t\,\mathrm{d}S_t + \beta_t\,\mathrm{d}B_t$, and the terminal value equals $g(S_T)$. □

The logical loop

self-financing replication $\iff$ Black–Scholes PDE $\iff$ discounted expectation.

Replication explains uniqueness and identifies the hedge. The expectation computes the PDE solution. Verification checks that both descriptions agree.

6 The European Call and Put

6.1 A truncated Gaussian identity

Let

$$\varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}, \qquad \mathrm{N}(x) = \int_{-\infty}^{x} \varphi(z)\,\mathrm{d}z.$$

Lemma 6.1 (Exponential tilting of a Gaussian tail)**.** For $a, b \in \mathbb{R}$ and $Z \sim N(0,1)$,

$$\boxed{\mathbb{E}\left[e^{aZ}\mathbf{1}_{\{Z>b\}}\right] = e^{a^2/2}\mathrm{N}(a-b).}$$

Proof. Complete the square:

$$e^{az}\varphi(z) = e^{a^2/2}\varphi(z-a).$$

Integrating over (b,∞) and substituting $y = z - a$ gives the result. □

6.2 The exercise event produces d_-

For a call,

$$g(s) = (s-K)^+, \qquad \tau = T - t.$$

Under the auxiliary drift-r dynamics,

$$S_T^r = s\exp\left(\left(r - \frac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}Z\right).$$

The price splits as

$$C(t,s) = e^{-r\tau}\left(\mathbb{E}[S_T^r\mathbf{1}_{\{S_T^r>K\}}] - K\mathbb{P}(S_T^r > K)\right).$$

Now

$$S_T^r > K \iff Z > -d_-,$$

where

$$\boxed{d_- = \frac{\log(s/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.}$$

Therefore

$$\mathbb{P}(S_T^r > K) = \mathrm{N}(d_-).$$

6.3 The stock-weighted term produces d_+

Apply the truncated Gaussian lemma with $a = \sigma\sqrt{\tau}$ and $b = -d_-$:

$$\mathbb{E}\left[e^{\sigma\sqrt{\tau}Z}\mathbf{1}_{\{Z>-d_-\}}\right] = e^{\sigma^2\tau/2}\mathrm{N}(d_- + \sigma\sqrt{\tau}).$$

Define

$$\boxed{d_+ = d_- + \sigma\sqrt{\tau} = \frac{\log(s/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.}$$

It follows that

$$\mathbb{E}[S_T^r\mathbf{1}_{\{S_T^r>K\}}] = se^{r\tau}\mathrm{N}(d_+).$$

Theorem 6.2 (Black–Scholes call and put formulas)**.** For $\tau = T - t$,

$$\boxed{C(t,s) = s\mathrm{N}(d_+) - Ke^{-r\tau}\mathrm{N}(d_-),}$$

where

$$\boxed{d_\pm = \frac{\log(s/K) + r\tau}{\sigma\sqrt{\tau}} \pm \frac{1}{2}\sigma\sqrt{\tau}.}$$

The put price is

$$\boxed{P(t,s) = Ke^{-r\tau}\mathrm{N}(-d_-) - s\mathrm{N}(-d_+).}$$

Put formula. The terminal identity

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K$$

is replicated by one share minus a zero-coupon payment K. No arbitrage gives put–call parity:

$$C(t,s) - P(t,s) = s - Ke^{-r\tau}.$$

Insert the call formula and use $\mathrm{N}(-x) = 1 - \mathrm{N}(x)$. □

Do not interpret both normal terms as the same probability

$\mathrm{N}(d_-)$ is the exercise probability under the drift-r pricing dynamics. The factor $\mathrm{N}(d_+)$ comes from a stock-weighted truncated expectation. Its later interpretation requires a stock-numeraire measure.

Example — boundary and maturity checks

The formula satisfies

$$\begin{array}{lll} s \downarrow 0: & d_\pm \to -\infty, & C(t,s) \to 0, \\ s \uparrow \infty: & d_\pm \to +\infty, & C(t,s) \sim s - Ke^{-r\tau}, \\ \tau \downarrow 0: & & C(t,s) \to (s-K)^+. \end{array}$$

It also satisfies the no-arbitrage bounds

$$(s - Ke^{-r\tau})^+ \le C(t,s) \le s.$$

These checks catch sign errors in $r\tau$, discounting, and $d_\pm$.

Exercise — a cash-or-nothing digital

Find the price of the payoff

$$g(S_T) = \mathbf{1}_{\{S_T > K\}}.$$

Short solution

The payoff is one unit of cash exactly on the call exercise event. Therefore

$$\boxed{D(t,s) = e^{-r\tau}\mathrm{N}(d_-).}$$

The discontinuity at maturity produces a sharply concentrated delta near the strike as $\tau \downarrow 0$.

7 Greeks and the Replicating Strategy

7.1 One identity simplifies every derivative

For the call formula,

$$C = s\mathrm{N}(d_+) - Ke^{-r\tau}\mathrm{N}(d_-).$$

The normal densities satisfy

$$\boxed{s\varphi(d_+) = Ke^{-r\tau}\varphi(d_-).}$$

Indeed, $d_- = d_+ - \sigma\sqrt{\tau}$ and direct simplification gives

$$\frac{\varphi(d_-)}{\varphi(d_+)} = \exp\left(\frac{d_+^2 - d_-^2}{2}\right) = \frac{se^{r\tau}}{K}.$$

7.2 Delta and gamma

Because

$$\partial_s d_+ = \partial_s d_- = \frac{1}{s\sigma\sqrt{\tau}},$$

differentiating the call price gives

$$\begin{aligned} C_s &= \mathrm{N}(d_+) + s\varphi(d_+)\partial_s d_+ - Ke^{-r\tau}\varphi(d_-)\partial_s d_- \\ &= \mathrm{N}(d_+). \end{aligned}$$

Thus

$$\boxed{\Delta_C = C_s = \mathrm{N}(d_+).}$$

Differentiating again,

$$\boxed{\Gamma_C = C_{ss} = \frac{\varphi(d_+)}{s\sigma\sqrt{\tau}}.}$$

Hedging interpretation

Delta is the number of shares in the replicating portfolio. Gamma measures how quickly the delta changes. Large gamma makes continuous-time replication especially sensitive to discrete rebalancing, transaction costs, jumps, and volatility misspecification.

7.3 Vega, theta, and rho

Differentiating with respect to volatility and using the density identity gives

$$\boxed{\mathrm{Vega}_C = C_\sigma = s\varphi(d_+)\sqrt{\tau}.}$$

Vega is positive because the call payoff is convex.

Theta is most efficiently computed from the PDE:

$$C_t = rC - rsC_s - \frac{1}{2}\sigma^2 s^2 C_{ss}.$$

Substitution yields

$$\boxed{\Theta_C = C_t = -\frac{s\sigma\varphi(d_+)}{2\sqrt{\tau}} - rKe^{-r\tau}\mathrm{N}(d_-).}$$

Here theta means calendar-time sensitivity $\partial C/\partial t$.

Finally,

$$\partial_r d_+ = \partial_r d_- = \frac{\sqrt{\tau}}{\sigma}.$$

The density contributions cancel, giving

$$\boxed{\mathrm{Rho}_C = C_r = K\tau e^{-r\tau}\mathrm{N}(d_-).}$$

Greek	Call formula	Main role
Δ	$\mathrm{N}(d_+)$	stock hedge
Γ	$\varphi(d_+)/(s\sigma\sqrt{\tau})$	hedge curvature
Vega	$s\varphi(d_+)\sqrt{\tau}$	volatility sensitivity
Θ	$-s\sigma\varphi(d_+)/(2\sqrt{\tau}) - rKe^{-r\tau}\mathrm{N}(d_-)$	time decay
Rho	$K\tau e^{-r\tau}\mathrm{N}(d_-)$	rate sensitivity

Exercise — put Greeks from parity

Use

$$P = C - s + Ke^{-r\tau}$$

to derive the five put Greeks.

Short solution

$$\Delta_P = \mathrm{N}(d_+) - 1 = -\mathrm{N}(-d_+),$$
$$\Gamma_P = \Gamma_C, \qquad \mathrm{Vega}_P = \mathrm{Vega}_C,$$
$$\Theta_P = -\frac{s\sigma\varphi(d_+)}{2\sqrt{\tau}} + rKe^{-r\tau}\mathrm{N}(-d_-),$$
$$\mathrm{Rho}_P = -K\tau e^{-r\tau}\mathrm{N}(-d_-).$$

Greeks are local model derivatives

A Greek measures an infinitesimal sensitivity while holding the rest of the model fixed. It is not a bound on finite losses. Near maturity or around a payoff kink, higher-order terms and discrete trading can dominate a first-order approximation.

8 Discrete Hedging and Model Limitations

8.1 The continuous hedge becomes a grid strategy

On a grid $0 = t_0 < \cdots < t_N = T$, a practical delta hedge uses

$$\Delta_n = C_s(t_n, S_{t_n})$$

over $(t_n, t_{n+1}]$. In discounted units,

$$\bar{S}_t = e^{-rt}S_t, \qquad \bar{X}_t = e^{-rt}X_t.$$

The self-financing recursion is

$$\boxed{\bar{X}_{t_{n+1}} = \bar{X}_{t_n} + \Delta_n(\bar{S}_{t_{n+1}} - \bar{S}_{t_n}).}$$

Starting from the Black–Scholes price,

$$\boxed{\bar{X}_T^N = C(0, S_0) + \sum_{n=0}^{N-1} \Delta_n(\bar{S}_{t_{n+1}} - \bar{S}_{t_n}).}$$

The terminal hedging error is

$$\mathrm{PL}_T^N = X_T^N - (S_T - K)^+.$$

In the ideal model, the error converges to zero as the trading mesh tends to zero under suitable conditions.

Why real hedging errors remain

- Trading occurs at discrete times and incurs costs.
- Volatility is not known and is not constant.
- Prices can jump and liquidity can disappear.
- Funding, collateral, dividends, and borrowing rates differ.
- The model may omit relevant state variables.

The Black–Scholes hedge is exact only inside its assumptions.

Exercise — leading delta–gamma hedging error

Over a short interval Δt, freeze delta at the beginning of the interval. Use a second-order Taylor expansion to identify the term that a delta-only hedge does not remove.

Short solution

For a smooth option value,

$$\Delta C \approx C_t\Delta t + C_s\Delta S + \frac{1}{2}C_{ss}(\Delta S)^2.$$

After subtracting the frozen-delta stock gain $C_s \Delta S$, the leading unhedged price-move term is

$$\boxed{\frac{1}{2}\Gamma(\Delta S)^2.}$$

The PDE balances its model-implied mean with theta and financing in continuous time. Realized squared moves need not match the model-implied variance over a discrete interval.

9 Guided Practice and Common Traps

9.1 A complete replication drill

Exercise — a smooth European payoff

Let $g(s) = s^2$.

1. Use the general expectation formula to find $p(t, s)$.
2. Compute the replicating holdings Δ_t, β_t.
3. Verify the PDE directly.

Short solution

The power-payoff formula with $m = 2$ gives

$$p(t, s) = s^2 e^{(r+\sigma^2)(T-t)}.$$

Hence

$$\Delta_t = 2S_t e^{(r+\sigma^2)(T-t)}$$

and

$$\beta_t = -\frac{S_t^2 e^{(r+\sigma^2)(T-t)}}{B_t}.$$

Direct substitution of

$$p_t = -(r + \sigma^2)p, \qquad p_s = 2p/s, \qquad p_{ss} = 2p/s^2$$

into the Black–Scholes PDE gives zero.

9.2 Error diagnosis

Exercise — correct four statements

Correct the following claims:

1. $X_t = \Delta_t S_t + \beta_t B_t$ automatically implies self-financing.
2. Rebalancing changes portfolio wealth.
3. A delta-hedged claim always earns r.
4. The call exercise probability is $\mathrm{N}(d_+)$.

Short solution

1. The value identity must be supplemented by $\mathrm{d}X = \Delta\,\mathrm{d}S + \beta\,\mathrm{d}B$.
2. Self-financing rebalancing changes composition but not current wealth.
3. The entire self-financing locally riskless portfolio earns r under no arbitrage; the option alone remains risky.
4. Under the pricing dynamics, the exercise probability is $\mathrm{N}(d_-)$.

9.3 End-of-lecture checklist

Ten checks for a replication argument

1. State the stock and bank-account dynamics.
2. Separate model assumptions from market assumptions.
3. Specify the payoff and terminal condition.
4. Define holdings and portfolio value.
5. Impose predictability and self-financing.
6. Apply Itô's formula to the candidate value.
7. Match diffusion to identify delta.
8. Match drift to derive the PDE.
9. Check regularity, boundary behavior, and integrability.
10. Verify that the strategy is self-financing and matches the payoff.

Final synthesis

- Self-financing is a cash-flow constraint, not merely a portfolio identity.
- Completeness allows Brownian risk to be replicated; no arbitrage makes the replication price unique.
- The physical drift cancels because the option and hedge carry the same stock exposure.
- Logarithmic coordinates turn the Black–Scholes PDE into a heat equation.
- The d_- term describes exercise under pricing dynamics; d_+ comes from a stock-weighted Gaussian expectation.
- Delta is the hedge, gamma measures its instability, and the remaining Greeks diagnose parameter and carry sensitivity.